

Modelling approach

For a baseline model we started with linear regression as it provides simple, easily interpretable coefficients. The diagnostic checks indicated that the model assumptions were reasonably fulfilled.

Then for checking non-linear patterns or complex feature interactions in the data we fitted a random forest model.

They showed similar predictive accuracy suggesting that the simple and more complex model capture the same underlying relationships in the dataset.

Business Value of Price Prediction for Airbnb

Accurate price prediction models can strengthen Airbnb's business. By drawing on core features such as room type, location, and availability, Airbnb can make use of this approach in several ways. Our recommendations:

Support hosts with pricing guidance. Recommend competitive nightly rates that increase bookings and reduce the risk of under- or over-pricing.

Forecast demand across markets. Identify neighbourhoods and seasons where demand will rise, allowing targeted marketing and host engagement.

Enhance guest trust. Reduce pricing inconsistencies so similar listings in the same area are priced more fairly, improving guest satisfaction.

Strategic insights. Factors with most influence on pricing help to guide investment decisions and regional growth priorities.

In summary, a reliable price prediction model equips Airbnb to align host pricing with market dynamics, build guest trust, and unlock new revenue opportunities.

Suggested visualisations

1. Model Evaluation: Predicted vs Actual Prices

Scatterplot of y_{test} vs y_{pred} with a 45° reference line.

2. Feature Importance: Linear Regression Coefficients (and/or RF coeffs)

Horizontal bar chart showing top 5 features.